

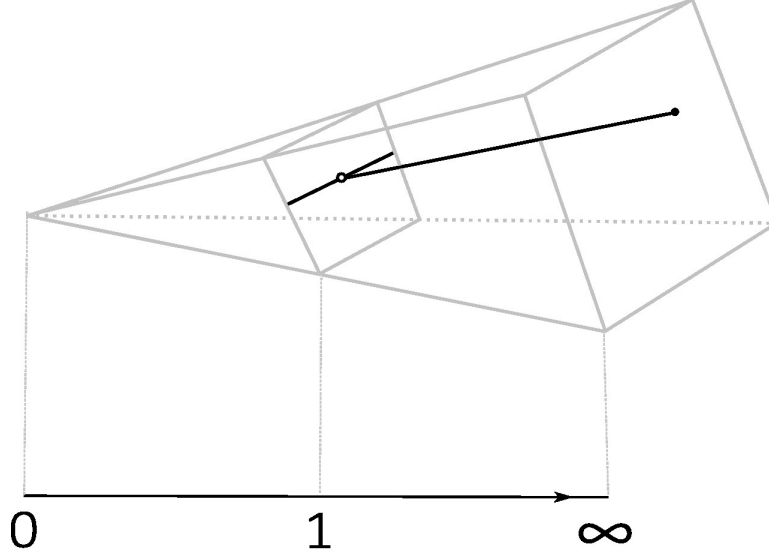


FIGURE 1. This picture represents the image under VAL of a line tangent to the quadric $Q(\mathbb{K})$.

a matrix power series $A(t)$ with $\det(A(t))$ not identically zero power series. Normalize to have $\det(A(t)) = 1$. Then $A(t) = t^\gamma cB + o(t^\gamma)$ with $\gamma > 0$, and $c \in \mathbb{C}^*$. Make change of variables in $A(t)$ giving $\tilde{A}(t)$ by replacing each t^β by $\exp(-\beta\gamma^{-1}\sigma)t^{\alpha\beta\gamma^{-1}}$, where $\sigma = \log(c)$. This comes from the necessity to obtain the image of the term ct^γ in the desired form as t^α , and it is an automorphism of $\mathbb{K}$ fixing $\mathbb{C}$. Then $\tilde{A}(t) = t^\alpha B + o(t^\alpha) \in V(\mathbb{K}) \cap PSL_2(\mathbb{K})$ and $\text{VAL}(\tilde{A}(t)) = (\alpha, [B]_{\mathbb{R}^*})$. $\square$

3. EXAMPLES

Example 1. We describe the image under VAL of a particular line L tangent to the quadric $Q(\mathbb{K})$. Consider $L \subset \mathbb{K}P^3$ parametrized by

$$\mathbb{K}P^1 \ni [z : w] \mapsto Z(t) = \begin{bmatrix} tw & z \\ 0 & t^{-1}w \end{bmatrix} \in \overline{PSL_2} \mathbb{K} = \mathbb{K}P^3,$$

$t \in \mathbb{K}$ being the uniformizer. The point where L intersects the quadric $Q(\mathbb{K})$ is $B_\infty = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ corresponding to $w = 0$.

For all other values of $[z : 1] \in \mathbb{K}P^1$ and $z = ct^\gamma + o(t^\gamma)$, $c \in \mathbb{C}^*$, we have three cases: $\gamma > 1$, $\gamma = 1$ and $\gamma < 1$. Look for the limit at ∞ for z represented by decreasing exponents.

In the first case, the dominant term of $Z(t)$ is $\begin{bmatrix} 0 & ct^\gamma \\ 0 & 0 \end{bmatrix}$. Projected by VAL on the cone picture of $\overline{PSL_2} \mathbb{C} = \mathbb{C}P^3$, it is a ray over B_∞ , each point with an S^1 in the circle bundle $\mathcal{S}$.

In the second case, the dominant term is $\begin{bmatrix} t & ct \\ 0 & 0 \end{bmatrix}$, giving a section of the $\mathcal{S}$ bundle over $Q(\mathbb{C})$ with points $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ excluded.